Chapter 9: Linear Regression

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November 3, 2026

CSAI2121 Probability and Statistics
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Learning Objectives

By the end of this lecture, you should be able to:

- Formulate a **simple** and a **multiple** linear regression model
- Define fitted values, residuals, and the sum of squared errors
- Derive the least-squares estimates for one input feature
- Write a regression model in matrix form $\mathbf{y} = X\mathbf{w} + \boldsymbol{\varepsilon}$
- Explain least squares as an **orthogonal projection**
- Derive and solve the **normal equations**
- Explain why OLS is the MLE under Gaussian noise
- Make predictions and evaluate a model using MSE, RMSE, and R^2

Slide Labels

Slides marked **[Extension]** are for further reading and are not examined.

From Maximum Likelihood to Regression

Recap: From a Gaussian Mean to Regression

Last week, every observation had the **same** mean:

$$X_i \sim N(\mu, \sigma^2), \qquad \widehat{\mu}_{\mathrm{MLE}} = \overline{x}.$$

In many AI problems, the mean depends on an **input feature**:

$$Y_i \sim N(\mu_i, \sigma^2), \qquad \mu_i = w_0 + w_1 x_i.$$

Key Idea

Linear regression is Gaussian MLE in which the mean is a linear function of the inputs.

- Last week: estimate one number μ.
- This week: estimate a weight vector $\mathbf{w}$.

What Is a Regression Problem?

A **regression** problem predicts a **numerical** target y from input features $\mathbf{x}$.

Input features $\mathbf{x}$	Target y
Floor area, age of building	House price
Study hours, attendance	Examination score
Advertising budget	Monthly sales
Sensor readings	Temperature
Image features	Estimated age of a person

- **Training data**: $(\mathbf{x}_1, y_1), \ldots, (\mathbf{x}_n, y_n)$.
- **Goal**: learn a function f such that $f(\mathbf{x}) \approx y$.
- **Classification** predicts a category; **regression** predicts a number.

Simple Linear Regression Model

With one input feature x:

$$\boxed{y_i = w_0 + w_1 x_i + \varepsilon_i, \qquad i = 1, \ldots, n}$$

- w_0: **intercept**, the predicted value when $x = 0$
- w_1: **slope**, the change in predicted y when x increases by one unit
- ε_i: **noise** or error term, i.e. everything not explained by x

What Is "Linear"?

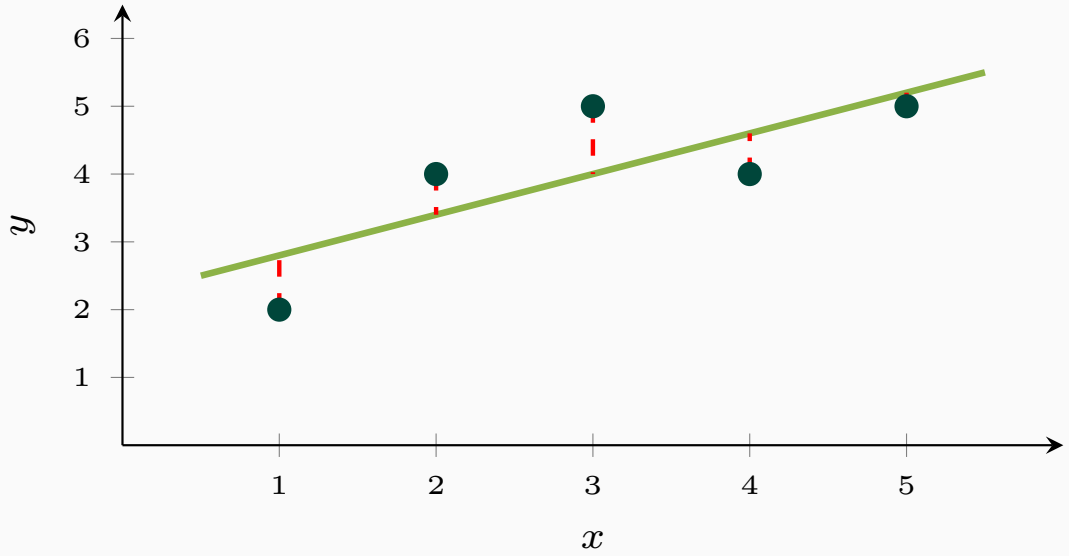
The model is linear in the **parameters** w_0, w_1. The term "linear" refers to how the weights enter the model.

The weights w_0, w_1 are unknown and must be estimated from data.

Fitted Values and Residuals

For candidate weights w_0, w_1:

fitted value: $\widehat{y}_i = w_0 + w_1 x_i$, residual: $e_i = y_i - \widehat{y}_i$.



- Dots: observed data; green line: fitted line; red dashes: residuals.
- A good line makes the residuals small overall.

The Least-Squares Criterion

Sum of Squared Errors

$$\mathrm{SSE}(w_0, w_1) = \sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n} \big(y_i - w_0 - w_1 x_i\big)^2.$$

Ordinary Least Squares (OLS) chooses the weights that minimize SSE.

Why square the residuals?

- Positive and negative residuals cannot cancel each other.
- Large errors are penalized more heavily than small errors.
- The objective is smooth, so we can use differentiation.
- It corresponds to a Gaussian noise model (Segment D).

Connection to Chapter 8

Maximizing a Gaussian likelihood was equivalent to minimizing squared errors.

Least Squares with One Feature

Warm-up: A Line Through the Origin

Consider first the model without intercept: $y_i = wx_i + \varepsilon_i$.

The objective has **one** variable:

$$\mathrm{SSE}(w) = \sum_{i=1}^{n} (y_i - wx_i)^2.$$

Single-variable differentiation gives:

$$\frac{d\ \mathrm{SSE}}{dw} = -2 \sum_{i=1}^{n} x_i (y_i - wx_i) = 0.$$

Solving:

$$\boxed{\widehat{w} = \frac{\sum_{i=1}^{n} x_i y_i}{\sum_{i=1}^{n} x_i^2}}$$

Since $\dfrac{d^2\ \mathrm{SSE}}{dw^2} = 2 \sum x_i^2 > 0$, this is a minimum.

Adding an Intercept: Two Partial Derivatives

Now SSE depends on two parameters w_0 and w_1.

As in Chapter 8, differentiate with respect to one parameter while holding the other fixed.

$$\frac{\partial \text{ SSE}}{\partial w_0} = -2\sum_{i=1}^{n}(y_i - w_0 - w_1 x_i) = 0$$

$$\frac{\partial \text{ SSE}}{\partial w_1} = -2\sum_{i=1}^{n} x_i(y_i - w_0 - w_1 x_i) = 0$$

Two Equations, Two Unknowns

$$\text{(E1)} \quad \sum_{i=1}^{n} e_i = 0, \qquad \text{(E2)} \quad \sum_{i=1}^{n} x_i e_i = 0.$$

Each derivative is an ordinary single-variable derivative.

Solving the Two Equations

From (E1): $\sum y_i - nw_0 - w_1 \sum x_i = 0$, so:

$$w_0 = \overline{y} - w_1 \overline{x}.$$

Substitute into (E2):

$$\sum_{i=1}^{n} x_i \Big[(y_i - \overline{y}) - w_1 (x_i - \overline{x}) \Big] = 0.$$

Since $\sum (y_i - \overline{y}) = 0$ and $\sum (x_i - \overline{x}) = 0$, we may replace x_i by $x_i - \overline{x}$:

$$\sum_{i=1}^{n} (x_i - \overline{x})(y_i - \overline{y}) = w_1 \sum_{i=1}^{n} (x_i - \overline{x})^2.$$

Define:

$$S_{xy} = \sum_{i=1}^{n} (x_i - \overline{x})(y_i - \overline{y}), \qquad S_{xx} = \sum_{i=1}^{n} (x_i - \overline{x})^2.$$

OLS Formulas for Simple Linear Regression

Result

$$\boxed{\widehat{w}_1 = \frac{S_{xy}}{S_{xx}}, \qquad \widehat{w}_0 = \overline{y} - \widehat{w}_1 \overline{x}}$$

- $\widehat{w}_1$ has the same sign as the sample covariance between x and y.
- Since $\widehat{w}_0 = \overline{y} - \widehat{w}_1 \overline{x}$, the fitted line always passes through $(\overline{x}, \overline{y})$.
- We need $S_{xx} > 0$, i.e. the x_i are not all equal.

Connection to Chapter 6

$$\widehat{w}_1 = \frac{\text{sample covariance of } x, y}{\text{sample variance of } x}.$$

Worked Example: Computing S_{xx} and S_{xy}

Data: $(1,2),\ (2,4),\ (3,5),\ (4,4),\ (5,5)$, so $\overline{x} = 3,\ \overline{y} = 4$.

x_i	y_i	$x_i - \overline{x}$	$y_i - \overline{y}$	$(x_i - \overline{x})(y_i - \overline{y})$	$(x_i - \overline{x})^2$
1	2	-2	-2	4	4
2	4	-1	0	0	1
3	5	0	1	0	0
4	4	1	0	0	1
5	5	2	1	2	4
15	20	0	0	$S_{xy} = 6$	$S_{xx} = 10$

Therefore:

$$\widehat{w}_1 = \frac{6}{10} = 0.6, \qquad \widehat{w}_0 = 4 - 0.6 \times 3 = 2.2.$$

Fitted Line

$$\widehat{y} = 2.2 + 0.6x.$$

Worked Example: Fitted Values and Residuals

Using $\widehat{y} = 2.2 + 0.6x$:

x_i	y_i	$\widehat{y}_i$	$e_i = y_i - \widehat{y}_i$	e_i^2
1	2	2.8	-0.8	0.64
2	4	3.4	0.6	0.36
3	5	4.0	1.0	1.00
4	4	4.6	-0.6	0.36
5	5	5.2	-0.2	0.04
			$\sum e_i = 0$	$\text{SSE} = 2.4$

Interpretation

No other straight line gives a sum of squared residuals smaller than 2.4 for these data.

Two Key Properties of OLS Residuals

Equations (E1) and (E2) say that, at the OLS solution:

$$\sum_{i=1}^{n} e_i = 0, \qquad \sum_{i=1}^{n} x_i e_i = 0.$$

Check with the example residuals $-0.8,\ 0.6,\ 1.0,\ -0.6,\ -0.2$:

$$\sum e_i = -0.8 + 0.6 + 1.0 - 0.6 - 0.2 = 0,$$

$$\sum x_i e_i = -0.8 + 1.2 + 3.0 - 2.4 - 1.0 = 0.$$

Vector Interpretation

Let $\mathbf{1} = (1, \ldots, 1)^T$, $\mathbf{x} = (x_1, \ldots, x_n)^T$, $\mathbf{e} = (e_1, \ldots, e_n)^T$:

$$\mathbf{1}^T \mathbf{e} = 0, \qquad \mathbf{x}^T \mathbf{e} = 0.$$

The residual vector is **orthogonal** to both $\mathbf{1}$ and $\mathbf{x}$.

Matrix Formulation and Normal Equations

Multiple Linear Regression

With d input features:

$$y_i = w_0 + w_1 x_{i1} + w_2 x_{i2} + \cdots + w_d x_{id} + \varepsilon_i.$$

Add a constant feature $x_{i0} = 1$ and write:

$$\mathbf{x}_i = \begin{bmatrix} 1 \\ x_{i1} \\ \vdots \\ x_{id} \end{bmatrix}, \qquad \mathbf{w} = \begin{bmatrix} w_0 \\ w_1 \\ \vdots \\ w_d \end{bmatrix}.$$

Then:

$$\boxed{y_i = \mathbf{x}_i^T \mathbf{w} + \varepsilon_i}$$

Writing a separate formula for every weight becomes impractical, so we use matrices instead.

The Design Matrix

Stack the n observations as rows:

$$\underbrace{\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}}_{\mathbf{y}\ (n\times 1)} = \underbrace{\begin{bmatrix} 1 & x_{11} & \cdots & x_{1d} \\ 1 & x_{21} & \cdots & x_{2d} \\ \vdots & \vdots & & \vdots \\ 1 & x_{n1} & \cdots & x_{nd} \end{bmatrix}}_{X\ (n\times(d+1))} \underbrace{\begin{bmatrix} w_0 \\ w_1 \\ \vdots \\ w_d \end{bmatrix}}_{\mathbf{w}} + \underbrace{\begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{bmatrix}}_{\boldsymbol{\varepsilon}}$$

Matrix Model

$$\boxed{\mathbf{y} = X\mathbf{w} + \boldsymbol{\varepsilon}}$$

- Each **row** of X is one observation.
- Each **column** of X is one feature (the first column is all ones).

Example: Design Matrix for Our Data

For the data $(1,2),(2,4),(3,5),(4,4),(5,5)$:

$$\mathbf{y} = \begin{bmatrix} 2 \\ 4 \\ 5 \\ 4 \\ 5 \end{bmatrix}, \qquad X = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \\ 1 & 5 \end{bmatrix}, \qquad \mathbf{w} = \begin{bmatrix} w_0 \\ w_1 \end{bmatrix}.$$

The vector of fitted values is:

$$\widehat{\mathbf{y}} = X\mathbf{w} = \begin{bmatrix} w_0 + w_1 \\ w_0 + 2w_1 \\ \vdots \\ w_0 + 5w_1 \end{bmatrix}.$$

The residual vector is $\mathbf{e} = \mathbf{y} - X\mathbf{w}$.

SSE in Matrix Form

The sum of squared residuals is the squared length of $\mathbf{e}$:

$$\mathrm{SSE}(\mathbf{w}) = \sum_{i=1}^{n} e_i^2 = \mathbf{e}^T\mathbf{e} = \|\mathbf{y} - X\mathbf{w}\|_2^2.$$

OLS Problem

$$\boxed{\widehat{\mathbf{w}} = \arg\min_{\mathbf{w}} \|\mathbf{y} - X\mathbf{w}\|_2^2}$$

Geometric Reading

Find the vector of the form $X\mathbf{w}$ that is **closest** to $\mathbf{y}$ in Euclidean distance.

We solve this using linear algebra rather than matrix calculus.

Geometric View: The Column Space of X

Write the columns of X as $\mathbf{c}_0, \mathbf{c}_1, \ldots, \mathbf{c}_d$. Then:

$$X\mathbf{w} = w_0\mathbf{c}_0 + w_1\mathbf{c}_1 + \cdots + w_d\mathbf{c}_d.$$

- Every possible fitted vector $X\mathbf{w}$ is a **linear combination** of the columns.
- The set of all such vectors is the **column space** $\mathrm{col}(X)$.
- Usually $\mathbf{y} \notin \mathrm{col}(X)$, so no $\mathbf{w}$ fits the data exactly.

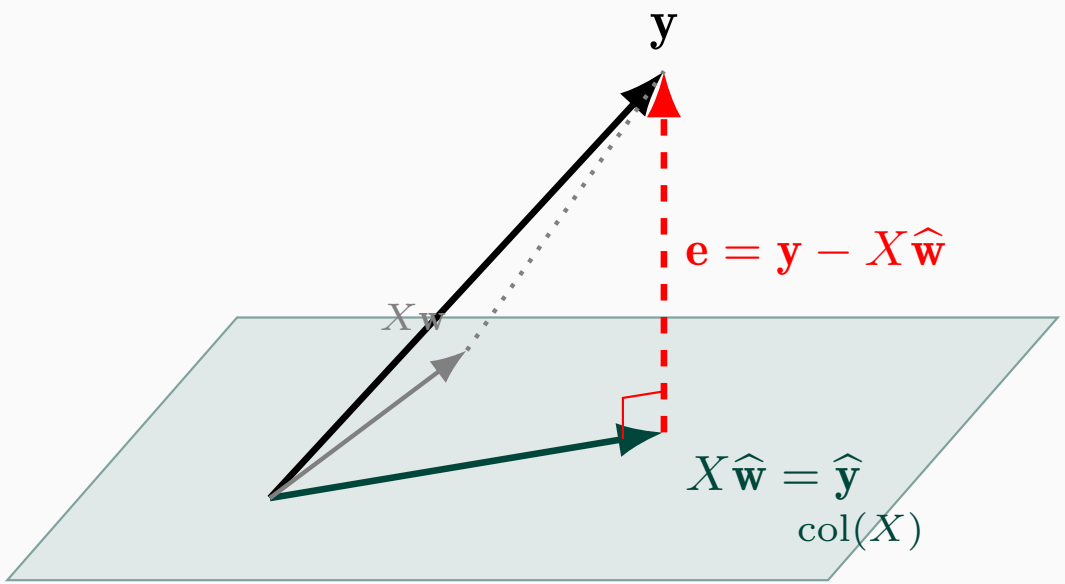
Least Squares Restated

Find the point in the subspace $\mathrm{col}(X)$ that is closest to $\mathbf{y}$.

Linear Algebra Fact

The closest point in a subspace is the **orthogonal projection** of $\mathbf{y}$ onto that subspace.

Least Squares as Orthogonal Projection



- Any other fitted vector $X\mathbf{w}$ (grey) is farther from $\mathbf{y}$.
- At the best fit, the residual $\mathbf{e}$ is **perpendicular** to the whole plane $\mathrm{col}(X)$.

From Orthogonality to the Normal Equations

The residual must be orthogonal to **every** column of X:

$$\mathbf{c}_j^T \left(\mathbf{y} - X\widehat{\mathbf{w}}\right) = 0, \qquad j = 0, 1, \ldots, d.$$

Stacking these $d+1$ equations:

$$X^T \left(\mathbf{y} - X\widehat{\mathbf{w}}\right) = \mathbf{0}.$$

Rearranging:

Normal Equations

$$\boxed{X^T X\, \widehat{\mathbf{w}} = X^T \mathbf{y}}$$

Why "Normal"?

"Normal" means *perpendicular*: the residual is normal to the column space. It is unrelated to the normal distribution.

The OLS Solution

If $X^T X$ is invertible:

$$\boxed{\widehat{\mathbf{w}}_{\text{OLS}} = (X^T X)^{-1} X^T \mathbf{y}}$$

- $X^T X$ is a $(d+1) \times (d+1)$ symmetric matrix.
- $X^T X$ is invertible exactly when the columns of X are **linearly independent**.
- The fitted values are $\widehat{\mathbf{y}} = X\widehat{\mathbf{w}}$.

Practical Computation

Software solves the linear system $X^T X \mathbf{w} = X^T \mathbf{y}$ (or uses QR decomposition) instead of explicitly computing the inverse, which is faster and numerically more stable.

Worked Example: Building the Normal Equations

With X from Slide 17:

$$X^T X = \begin{bmatrix} n & \sum x_i \\ \sum x_i & \sum x_i^2 \end{bmatrix} = \begin{bmatrix} 5 & 15 \\ 15 & 55 \end{bmatrix},$$

$$X^T \mathbf{y} = \begin{bmatrix} \sum y_i \\ \sum x_i y_i \end{bmatrix} = \begin{bmatrix} 20 \\ 66 \end{bmatrix}.$$

Here:

$$\sum x_i^2 = 1+4+9+16+25 = 55, \qquad \sum x_i y_i = 2+8+15+16+25 = 66.$$

Normal equations:

$$\begin{bmatrix} 5 & 15 \\ 15 & 55 \end{bmatrix} \begin{bmatrix} w_0 \\ w_1 \end{bmatrix} = \begin{bmatrix} 20 \\ 66 \end{bmatrix}.$$

Worked Example: Solving the Normal Equations

Determinant: $\det(X^T X) = 5 \times 55 - 15 \times 15 = 50.$

$$(X^T X)^{-1} = \frac{1}{50} \begin{bmatrix} 55 & -15 \\ -15 & 5 \end{bmatrix}.$$

Therefore:

$$\widehat{\mathbf{w}} = \frac{1}{50} \begin{bmatrix} 55 \times 20 - 15 \times 66 \\ -15 \times 20 + 5 \times 66 \end{bmatrix} = \frac{1}{50} \begin{bmatrix} 110 \\ 30 \end{bmatrix} = \begin{bmatrix} 2.2 \\ 0.6 \end{bmatrix}.$$

Consistency Check

The matrix method gives exactly the same line as the scalar formulas: $\widehat{y} = 2.2 + 0.6x$.

Matrix and Scalar Methods Agree

For simple regression, $X = [\,\mathbf{1}\ \ \mathbf{x}\,]$. The normal equations

$$X^T(\mathbf{y} - X\widehat{\mathbf{w}}) = \mathbf{0}$$

become:

$$\begin{bmatrix} \mathbf{1}^T\mathbf{e} \\ \mathbf{x}^T\mathbf{e} \end{bmatrix} = \begin{bmatrix} \sum e_i \\ \sum x_i e_i \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

These are exactly equations (E1) and (E2) from Slide 9.

Summary of the Two Viewpoints

- **Calculus view**: set the partial derivatives of SSE to zero.
- **Geometric view**: make the residual orthogonal to every column of X.

Both lead to the same normal equations.

When Is $X^T X$ Not Invertible?

$X^T X$ fails to be invertible when the columns of X are linearly dependent.

- **Duplicated information**: the same length recorded in both cm and inches, so one column is a multiple of another.
- **Too many features**: more parameters than observations, $d + 1 > n$.
- **Constant feature**: a feature that never changes duplicates the intercept column.

Consequences:

- The normal equations have infinitely many solutions.
- Even when $X^T X$ is *nearly* singular, estimates can be very unstable.

Remedies

Remove redundant features, collect more data, or add **regularization**, which is the topic of next week.

Hat Matrix and Gradient View [Extension]

Hat matrix. The fitted values are:

$$\widehat{\mathbf{y}} = X(X^T X)^{-1} X^T \mathbf{y} = H\mathbf{y}, \qquad H = X(X^T X)^{-1} X^T.$$

H is the projection matrix onto $\mathrm{col}(X)$; it "puts the hat on $\mathbf{y}$".

Gradient view. Using matrix calculus, one can show:

$$\nabla_{\mathbf{w}} \|\mathbf{y} - X\mathbf{w}\|_2^2 = -2X^T(\mathbf{y} - X\mathbf{w}).$$

Setting it to zero gives the same normal equations.

Not Examined

You are not required to perform matrix differentiation. The projection argument is sufficient.

Probabilistic Interpretation

The Gaussian Noise Model

Assume:

$$y_i = \mathbf{x}_i^T \mathbf{w} + \varepsilon_i, \qquad \varepsilon_i \overset{\text{i.i.d.}}{\sim} N(0, \sigma^2).$$

Equivalently:

$$\boxed{y_i \mid \mathbf{x}_i \sim N\left(\mathbf{x}_i^T \mathbf{w},\ \sigma^2\right)}$$

- The **mean** of y_i is the linear prediction $\mathbf{x}_i^T \mathbf{w}$.
- The **spread** around the line is the same for all observations.
- Compared with Chapter 8, the only change is $\mu \longrightarrow \mu_i = \mathbf{x}_i^T \mathbf{w}$.

Likelihood for Linear Regression

By independence:

$$L(\mathbf{w}, \sigma^2) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_i - \mathbf{x}_i^T \mathbf{w})^2}{2\sigma^2}\right).$$

Taking logarithms:

$$\ell(\mathbf{w}, \sigma^2) = -\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\underbrace{\sum_{i=1}^{n}(y_i - \mathbf{x}_i^T \mathbf{w})^2}_{\mathrm{SSE}(\mathbf{w})}$$

Observation

$\mathbf{w}$ appears only inside $\mathrm{SSE}(\mathbf{w})$, and it appears with a negative sign.

OLS Is the Gaussian MLE

For any fixed $\sigma^2 > 0$:

$$\arg\max_{\mathbf{w}} \ell(\mathbf{w}, \sigma^2) = \arg\min_{\mathbf{w}} \mathrm{SSE}(\mathbf{w}).$$

Main Result

$$\boxed{\widehat{\mathbf{w}}_{\mathrm{MLE}} = \widehat{\mathbf{w}}_{\mathrm{OLS}} = (X^T X)^{-1} X^T \mathbf{y}}$$

Repeating the variance derivation of Chapter 8 gives:

$$\widehat{\sigma}^2_{\mathrm{MLE}} = \frac{\mathrm{SSE}(\widehat{\mathbf{w}})}{n}.$$

Example: $\mathrm{SSE} = 2.4,\ n = 5 \ \Rightarrow\ \widehat{\sigma}^2_{\mathrm{MLE}} = 0.48.$

(The unbiased version divides by $n - (d+1)$: $2.4/3 = 0.8$.)

Model Assumptions

Assumption	Meaning
Linearity	The mean of y is linear in the weights
Independence	Noise terms of different observations are independent
Constant variance	The spread of ε_i does not depend on $\mathbf{x}_i$
Gaussian noise	$\varepsilon_i \sim N(0, \sigma^2)$
No perfect collinearity	Columns of X are linearly independent

Important Distinction

Computing OLS needs only linear independence of the columns. The Gaussian assumption is needed to interpret OLS as an MLE and to construct confidence intervals and tests (Weeks 11–12).

Prediction, Evaluation, and AI Applications

Making Predictions

For a new input $\mathbf{x}_{\text{new}}$:

$$\boxed{\widehat{y}_{\text{new}} = \mathbf{x}_{\text{new}}^T \widehat{\mathbf{w}}}$$

Example. With $\widehat{y} = 2.2 + 0.6x$ and $x_{\text{new}} = 6$:

$$\widehat{y}_{\text{new}} = \begin{bmatrix} 1 & 6 \end{bmatrix} \begin{bmatrix} 2.2 \\ 0.6 \end{bmatrix} = 2.2 + 3.6 = 5.8.$$

Caution: Extrapolation

Predictions far outside the range of the training inputs (here $1 \leq x \leq 5$) rely heavily on the assumption that the linear pattern continues.

Measuring the Quality of Fit

Common Metrics

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \widehat{y}_i)^2, \qquad \text{RMSE} = \sqrt{\text{MSE}},$$

$$R^2 = 1 - \frac{\text{SSE}}{\text{SST}}, \qquad \text{SST} = \sum_{i=1}^{n}(y_i - \overline{y})^2.$$

Example: $\text{SSE} = 2.4$ and $\text{SST} = 4 + 0 + 1 + 0 + 1 = 6$:

$$\text{MSE} = 0.48, \qquad \text{RMSE} \approx 0.693, \qquad R^2 = 1 - \frac{2.4}{6} = 0.6.$$

- RMSE is in the same unit as y.
- $R^2 = 0.6$: the model explains 60% of the variation in y.

Interpreting Coefficients in Multiple Regression

Suppose a fitted model for house prices is:

$$\widehat{\text{price}} = 50 + 3.2 \times \text{area} - 1.5 \times \text{age},$$

with price in 10^4 MOP, area in m^2, and age in years.

- $+3.2$: one extra m^2 is associated with 3.2×10^4 MOP higher predicted price, **holding age fixed**.
- -1.5: one extra year of age is associated with 1.5×10^4 MOP lower predicted price, **holding area fixed**.

Cautions

- Coefficients describe **associations**, not necessarily causal effects.
- Coefficient sizes depend on the units of the features.

Feature Engineering: Polynomial Regression

A curved relationship can still be handled by linear regression:

$$y_i = w_0 + w_1 x_i + w_2 x_i^2 + \varepsilon_i.$$

The model is **nonlinear in** x but **linear in** $\mathbf{w}$, so the same normal equations apply with:

$$X = \begin{bmatrix} 1 & x_1 & x_1^2 \\ 1 & x_2 & x_2^2 \\ \vdots & \vdots & \vdots \\ 1 & x_n & x_n^2 \end{bmatrix}.$$

Preview: Overfitting

Adding many polynomial terms always reduces the training SSE, but the model may fit noise and predict poorly on new data. Regularization (Week 10) addresses this.

Python Demonstration

```
import numpy as np

x = np.array([1, 2, 3, 4, 5], dtype=float)
y = np.array([2, 4, 5, 4, 5], dtype=float)

# Design matrix with a column of ones
X = np.column_stack([np.ones_like(x), x])

# Solve the normal equations  X^T X w = X^T y
w = np.linalg.solve(X.T @ X, X.T @ y)
print(w)                    # [2.2 0.6]

y_hat = X @ w
sse = np.sum((y - y_hat) ** 2)
print(sse)                  # 2.4

# Recommended in practice: least-squares solver
w_ls, *_ = np.linalg.lstsq(X, y, rcond=None)
```

AI Applications of Linear Regression

Application	Role of linear regression
Price and demand forecasting	Predict a numerical quantity from features
Baseline models	Simple, interpretable benchmark before deep learning
Neural networks	The last layer of a regression network is linear
Feature importance	Coefficients indicate associations with features
Linear probing	Evaluate learned representations with a linear model

Big Picture

Many neural networks trained with MSE loss can be viewed as fitting a Gaussian model with a learned, nonlinear mean function $f(\mathbf{x}; \mathbf{w})$.

Summary: Key Definitions and Formulas

Model: $\mathbf{y} = X\mathbf{w} + \boldsymbol{\varepsilon}, \qquad \boldsymbol{\varepsilon} \sim \text{i.i.d. } N(0, \sigma^2)$

Simple regression: $\widehat{w}_1 = \dfrac{S_{xy}}{S_{xx}}, \qquad \widehat{w}_0 = \overline{y} - \widehat{w}_1 \overline{x}$

Normal equations: $X^T X \, \widehat{\mathbf{w}} = X^T \mathbf{y}$

OLS solution: $\widehat{\mathbf{w}} = (X^T X)^{-1} X^T \mathbf{y}$

Noise variance: $\widehat{\sigma}^2_{\text{MLE}} = \dfrac{\text{SSE}}{n}$

Fit: $\text{MSE} = \dfrac{\text{SSE}}{n}, \qquad R^2 = 1 - \dfrac{\text{SSE}}{\text{SST}}$

Common Mistakes

- Forgetting the column of ones in the design matrix
- Writing $X\mathbf{w}$ with incompatible dimensions
- Confusing residuals $e_i = y_i - \widehat{y}_i$ with the unobservable noise ε_i
- Thinking "linear" means the model must be a straight line in x
- Writing $(X^T X)^{-1} = X^{-1}(X^T)^{-1}$, which is invalid when X is not square
- Assuming $X^T X$ is always invertible
- Interpreting regression coefficients as causal effects
- Trusting predictions far outside the training range
- Judging a model only by training R^2

Class Exercise

The observed data are:

$$(x_i, y_i): \quad (0,1),\ (1,3),\ (2,2),\ (3,5).$$

Problem 1: Scalar Formulas

1. Compute $\overline{x}$, $\overline{y}$, S_{xx}, and S_{xy}.
2. Find $\widehat{w}_1$ and $\widehat{w}_0$.

Problem 2: Matrix Method

1. Write X and $\mathbf{y}$, and compute $X^T X$ and $X^T \mathbf{y}$.
2. Solve the normal equations and check your answer to Problem 1.
3. Predict y at $x = 4$.
4. Compute the residuals, SSE, and $\widehat{\sigma}^2_{\mathrm{MLE}}$.

Work in pairs for 15 minutes, then discuss as a class.

Class Exercise: Solution to Problem 1

Means:

$$\overline{x} = \frac{0+1+2+3}{4} = 1.5, \qquad \overline{y} = \frac{1+3+2+5}{4} = 2.75.$$

Deviations: $x_i - \overline{x} = -1.5,\ -0.5,\ 0.5,\ 1.5$;
$y_i - \overline{y} = -1.75,\ 0.25,\ -0.75,\ 2.25.$

$$S_{xx} = 2.25 + 0.25 + 0.25 + 2.25 = 5,$$

$$S_{xy} = 2.625 - 0.125 - 0.375 + 3.375 = 5.5.$$

Therefore:

$$\boxed{\widehat{w}_1 = \frac{5.5}{5} = 1.1, \qquad \widehat{w}_0 = 2.75 - 1.1 \times 1.5 = 1.1}$$

Fitted line: $\widehat{y} = 1.1 + 1.1x.$

Class Exercise: Solution to Problem 2

$$X^T X = \begin{bmatrix} 4 & 6 \\ 6 & 14 \end{bmatrix}, \qquad X^T \mathbf{y} = \begin{bmatrix} 11 \\ 22 \end{bmatrix}, \qquad \det(X^T X) = 56 - 36 = 20.$$

$$\widehat{\mathbf{w}} = \frac{1}{20} \begin{bmatrix} 14 & -6 \\ -6 & 4 \end{bmatrix} \begin{bmatrix} 11 \\ 22 \end{bmatrix} = \frac{1}{20} \begin{bmatrix} 22 \\ 22 \end{bmatrix} = \begin{bmatrix} 1.1 \\ 1.1 \end{bmatrix}.$$

Prediction at $x = 4$: $\widehat{y} = 1.1 + 1.1 \times 4 = 5.5$.

Fitted values $1.1,\ 2.2,\ 3.3,\ 4.4$; residuals $-0.1,\ 0.8,\ -1.3,\ 0.6$.

$$\mathrm{SSE} = 0.01 + 0.64 + 1.69 + 0.36 = 2.7, \qquad \boxed{\widehat{\sigma}^2_{\mathrm{MLE}} = \frac{2.7}{4} = 0.675}$$

Check: $\sum e_i = 0$ and $\sum x_i e_i = 0 + 0.8 - 2.6 + 1.8 = 0$.

Preview of Next Week: MAP, Information Theory, and Markov Chains

- Today, we estimated $\mathbf{w}$ using the likelihood only (MLE $=$ OLS).
- Next week, we add a **prior** on the parameters:

$$p(\mathbf{w} \mid D) \propto p(D \mid \mathbf{w})\, p(\mathbf{w}).$$

- A Gaussian prior leads to **ridge regression**:

$$\widehat{\mathbf{w}} = (X^T X + \lambda I)^{-1} X^T \mathbf{y},$$

  which is always well defined for $\lambda > 0$.
- We will also introduce entropy, cross-entropy, and KL divergence, together with a short introduction to Markov Chains.

Main Connection

Regularization is not an ad hoc trick: it is MAP estimation with a prior distribution.